RABIES

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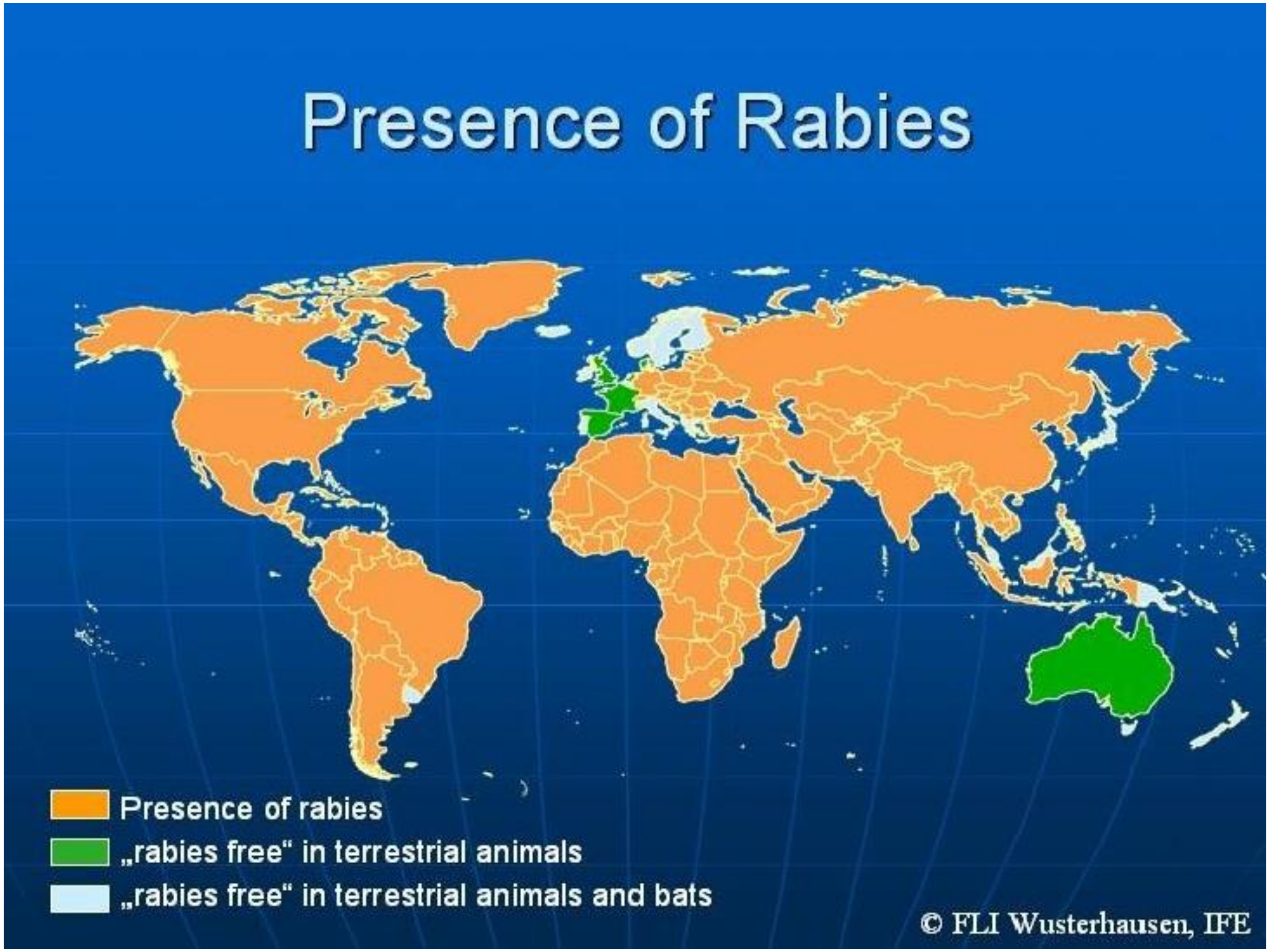
Synonyms- ***HYDROPHOBIA, LYSSA, JALATANKA***

Rabies is an acute,viral encephalomyelitis (inflammation of both brain and spinal cord), which affects all warm-blooded animals, including humans.

- Highly fatal disease.
- Mortality rate being close to 100%.
- Rabies occurs throughout the world.
- A few countries are free of the disease by virtue of their island status, or due to successful eradication programmes, or because of strict quarantine regulations.

- New Zealand and Australia never had the disease.
- Britain, Hawaii, Japan and Scandinavia are currently free of the disease.
- In United States, Canada, and Western Europe, rabies in dogs has been effectively controlled through vaccination.


Presence of Rabies
Presence of rabies
„rabies free“ in terrestrial animals
„rabies free“ in terrestrial animals and bats
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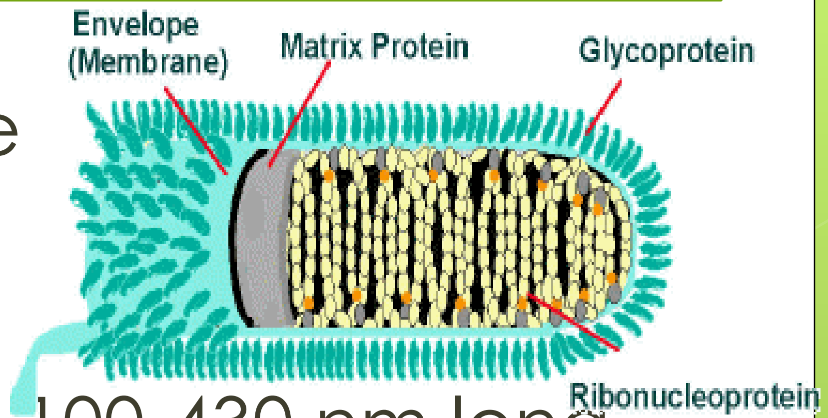
Etiology



- Family – Rhabdoviridae
- Genus – Lyssa virus
- Bullet-shaped
- 45-100 nm in diameter; 100-430 nm long.
- Enveloped, ss-RNA: - ve strand
- It encodes for 5 structural protein

(i) Nucleocapsid (N)
(ii) Phosphoprotein (P)
(iii) Matrix protein (M)
(iv) Glycoprotein (G)
(v) RNA dependent RNA polymerase (L)

- 7 different genotype of rabies virus reported.
- UV light and heat destroy virus quickly.
- Rabies virus can remain viable in carcass for several days at 20 °C.
- Storage at ultra low temperature (-30°c to - 80 °c) prolongs viral activity for years in untreated frozen samples.

Transmission

1. Bite of a rabid animal
2. By aerosol- mostly reported in extremely large colonies of cave dwelling bats.
3. By ingestion –

- Mostly seen in dogs eating flesh of rabies animals.
- By ingestion of raw milk of affected animal

(Satapathy *et.al*.2010

4. By transplacental – Reported in calf

Host and range

- Dog, and cat, act as a principal source of infection.
- Foxes, wolves, skunks, blood-sucking vampire bats, insectivorous and frugivorous (fruit-eating) bats, racoons, mongoose, and squirrels may provide the major source of infection in countries where domestic carnivore are well controlled.

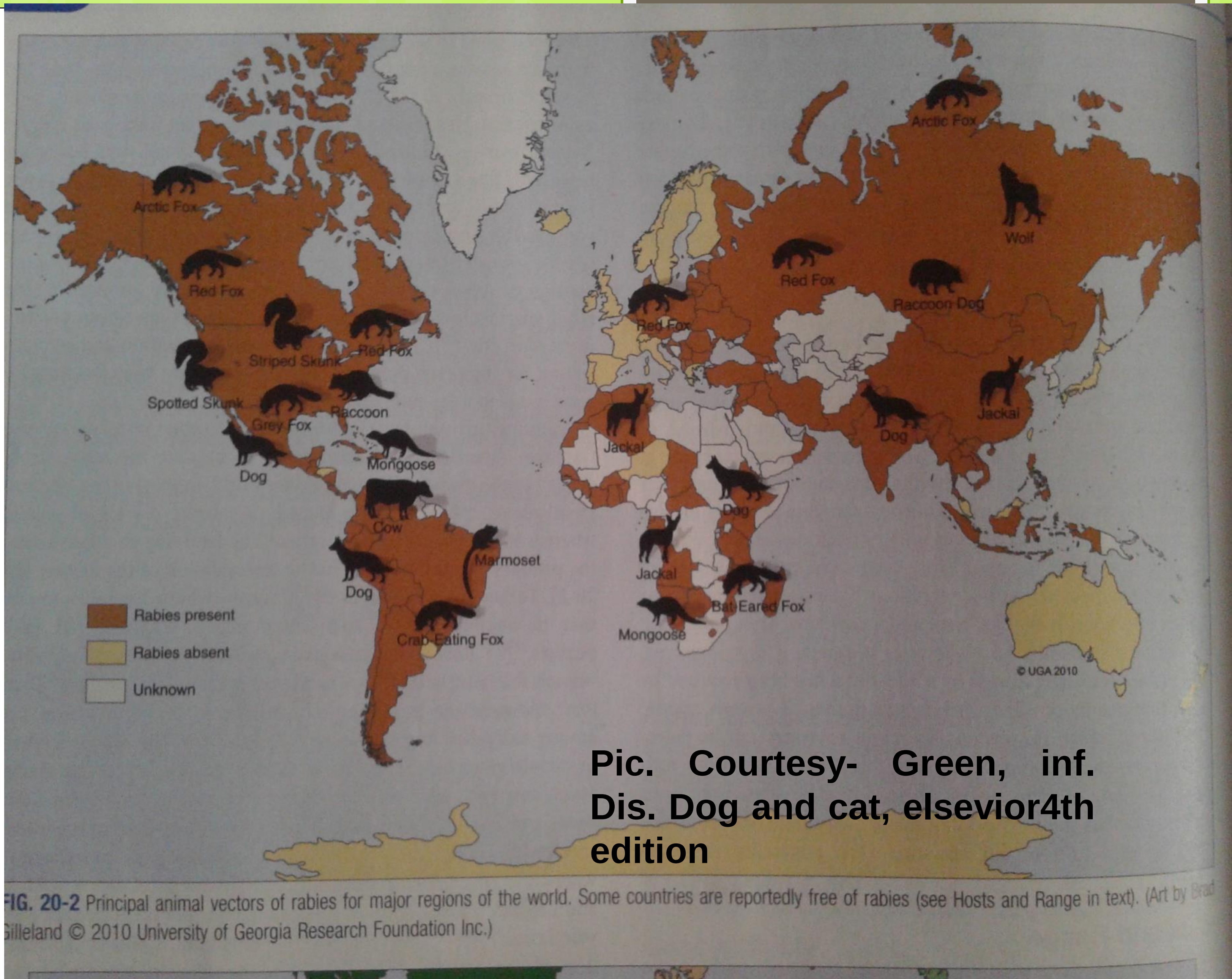


FIG. 20-2 Principal animal vectors of rabies for major regions of the world. Some countries are reportedly free of rabies (see Hosts and Range in text). (Art by Brad Gilleland © 2010 University of Georgia Research Foundation Inc.)

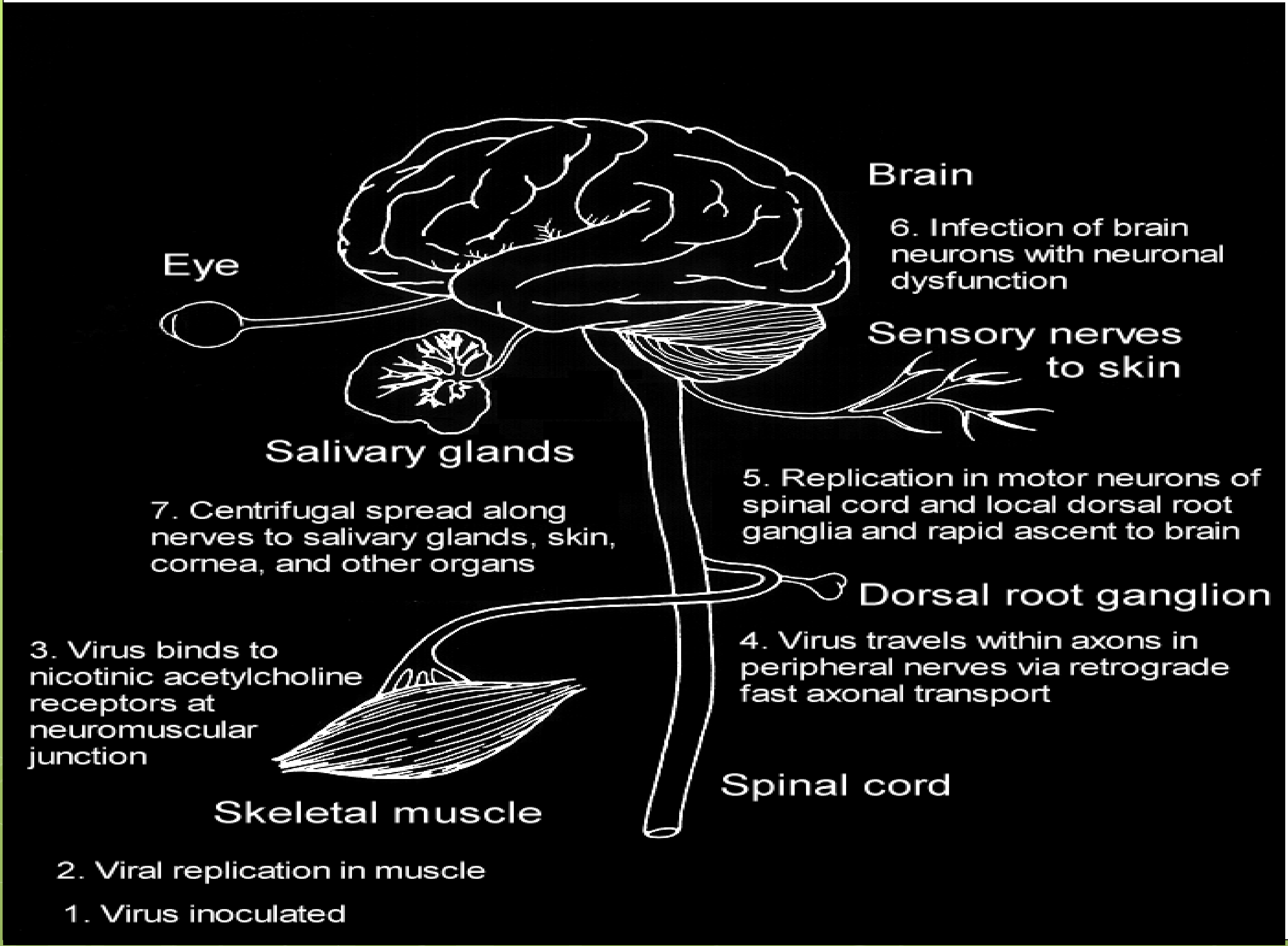
Pic. Courtesy- Green, inf. Dis. Dog and cat, elsevior4th edition

- ***The rabies virus is maintained by interrelated cycle-***
- **Urban cycle rabies**: dogs,cats
- **Sylvatic cycle rabies**: foxes, wolves, bats etc.
- Rabies in humans results from the bite of a rabid dog, cat, fox, or wolf.
- These animals also transmit the disease to cattle, horses, and sheep, which seldom spread it further.
- **Bats** are the important species in which **symptomless carriers** are known to occur. **Multiplication of the virus without invasion of the nervous system** is known to occur in fatty tissues in bats, and may be the basis of the 'reservoir' mechanism known to occur in this species.

TYPES OF RABIES VIRUS

	STREET VIRUS	FIXED VIRUS
	The virus recovered from naturally occurring cases of rabies is called street virus.	The virus which has a short , fixed and reproducible incubation period is called fixed virus.
SOURCES	It is naturally occurring virus . It is found in saliva of infected animal	It is prepared by repeated culture in brain of rabbit such that its IP is reduced and fixed.
FEATURES	1. It produce negri bodies 2. Incubation period is 20 to 60 days. 3. It is pathogenic for all mammals 4. Cannot be used for preparation of vaccine	1. It does not form negri bodies 2. Incubation period is constant between 4-6 days. 3. It can be pathogenic for humans under certain conditions. 4. It is used to prepare anti-rabies vaccine.

Pathogenesis



Pathogenesis of Rabies

Virus transmitted from saliva to muscle

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Virus travels from neuromuscular junction to peripheral nerves

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From PNS to CNS

↓

Infects brain, salivary gland

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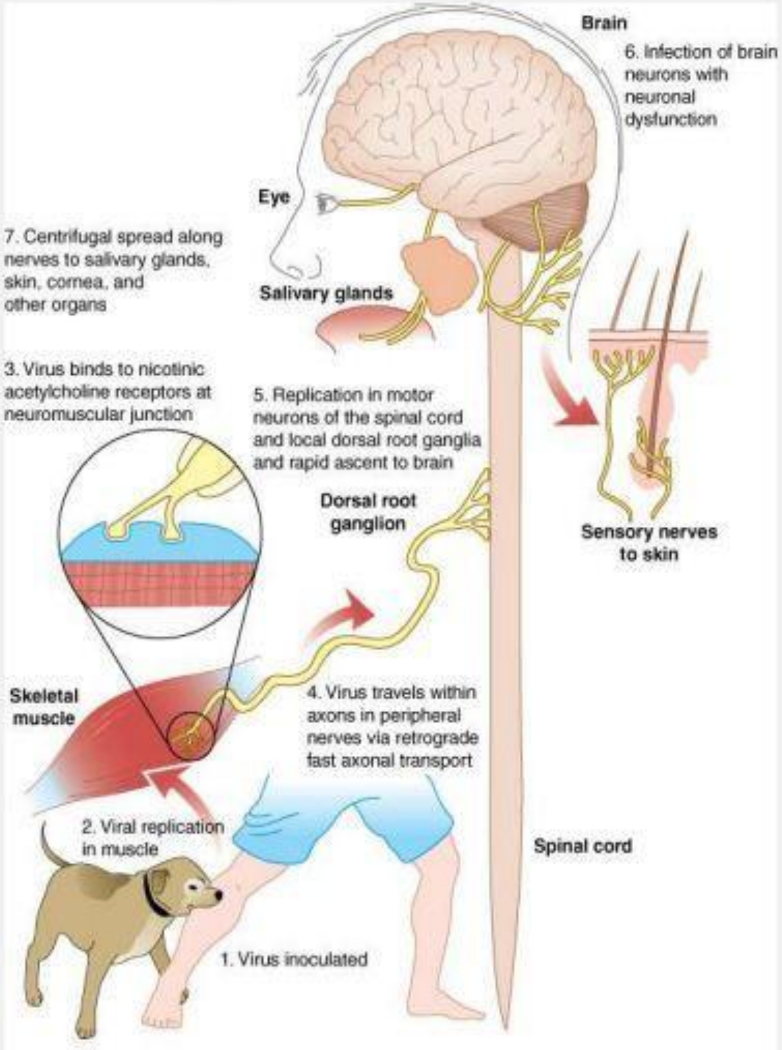
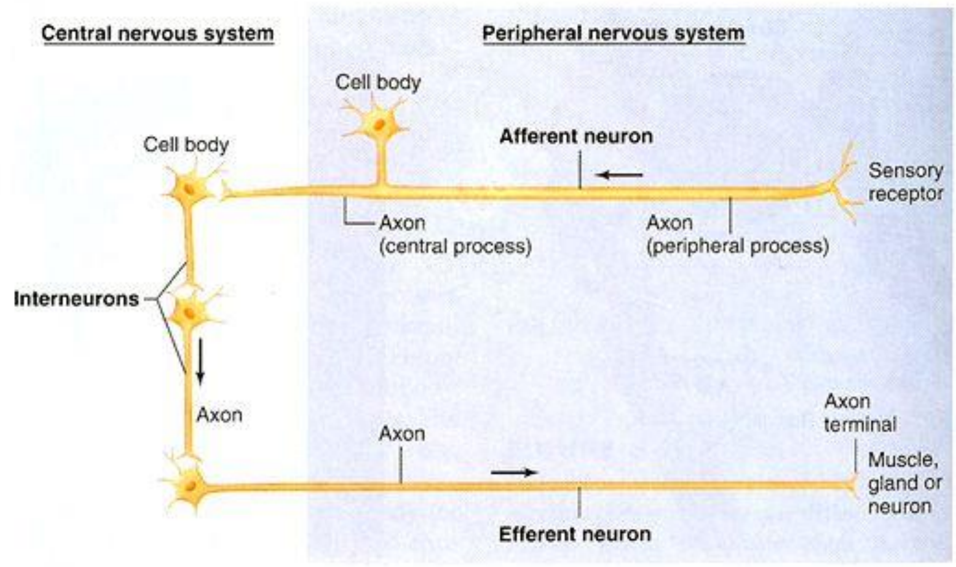
Infects other organs – respiratory tract

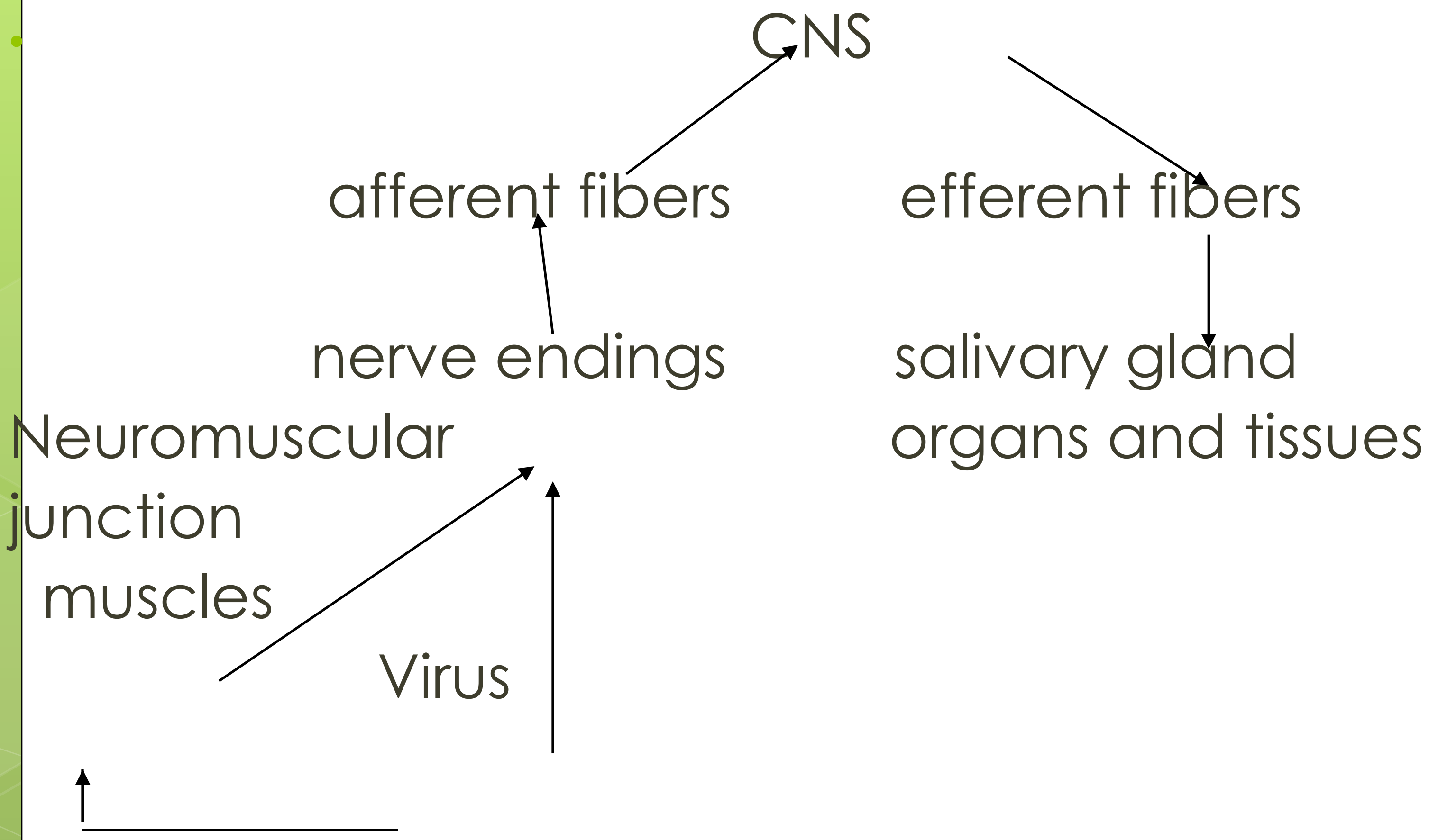


Source: Fauci AS, Kasper DL, Braunwald E, Hauser SL, Longo DL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine*, 17th Edition: http://www.accessmedicine.com

Mechanism

- The virus then travels along the nerves towards the central nervous system
- During this phase, the virus cannot be easily detected within the host




CNS
afferent fibers
efferent fibers
nerve endings
salivary gland
Neuromuscular
junction
organs and tissues
muscles
Virus

Symptoms

- **Incubation period-** vary from 10 days to 3 months in dogs
- There are two types – ***(1) Furious form*** ***(2) Dumb form***

✓ ***Furious form*** – can be divided into two stages

a) ***Stage of melancholy-*** Change in behavior, unusual violence and frenzy behavior, bite inanimate objects, alert condition, eat non-edible object e.g. stone, bone, grass pupils get dilated, altered facial expression.

b) ***Stage of excitement-*** Becomes aggressive due to excitability & irritability, hide in dark place due to photophobia, hydrophobia, champing of jaw & dribbling of saliva, lick their Genital & show sign of heat.

- At the end, the lower jaw will hang, head will drop down, ascending paralysis, coma & death.

➢ Dumb form:–

- It is also known as PARALYTIC FORM .
- In this form, paralysis of lower jaw, tongue, larynx & hind quarters.
- Dog cannot bite, saliva remain infective.
- Peculiar voice is produced known as HOWL. (deep cry)
- Dog used to seek solitude and appear sluggish & morose.
- Lasts for 1-7 days Coma and finally death..

FIG. 20-9 Dog with paralytic stage of rabies in sternal recumbency with torticollis. (Courtesy CDC, Atlanta.)

FIG. 20-10 Dog with rabies. Note open jaw and visible tongue with excessive salivary secretions resulting from the inability to swallow. (Courtesy CDC, Atlanta.)

Dumb form

Furious form

➢ ***Sign in horse*** -

- ✓ Weakness and lameness is initial sign.
- ✓ may show furious form & become virulent & uncontrollable fall & roll on the ground
- ✓ Shows signs of colic
- ✓ chew foreign object drooling of saliva & convulsion, sexual excitement

Cattle

- Rabies in cattle follows the same general pattern.
- Those with the furious form are dangerous, attacking and pursuing man and other animals.
- Lactation stops abruptly in dairy cattle.
- **Dribbling of urine (frequent urination)**
- **A most typical clinical sign in cattle is a characteristic bellowing (making a deep loud cry). This may continue intermittently until shortly before death.**

Rabies in wild animals

Symptoms

- The incubation period is usually two to three weeks.
- Different species show different signs of the disease. Expect variations even within the same species, because few animals show all of the signs of rabies. Some signs are subtle and easily missed.
- unprovoked aggression ("furious" rabies). Some animals may attack anything that moves, or even inanimate objects.
- unusual friendliness ("dumb" rabies).
- animal may stumble, fall, appear disoriented or uncoordinated, or wander aimlessly.

- Paralysis, often beginning in the hind legs or throat. Paralysis of the throat muscles can cause the animal to bark, whine, drool, choke, or froth at the mouth.
- Vocalizations ranging from chattering to shrill screams.
- Nocturnal animals may become unusually active during the day
- Raccoons walk as if they're on very hot pavement.
- Skunks, raccoons, foxes, and dogs usually display furious rabies.
- Bats often display dumb rabies, and may be found on the ground, unable to fly.

Clinical Course of Rabies in Humans

Exposure → First symptom → First Neurological sign → Onset of coma → Death

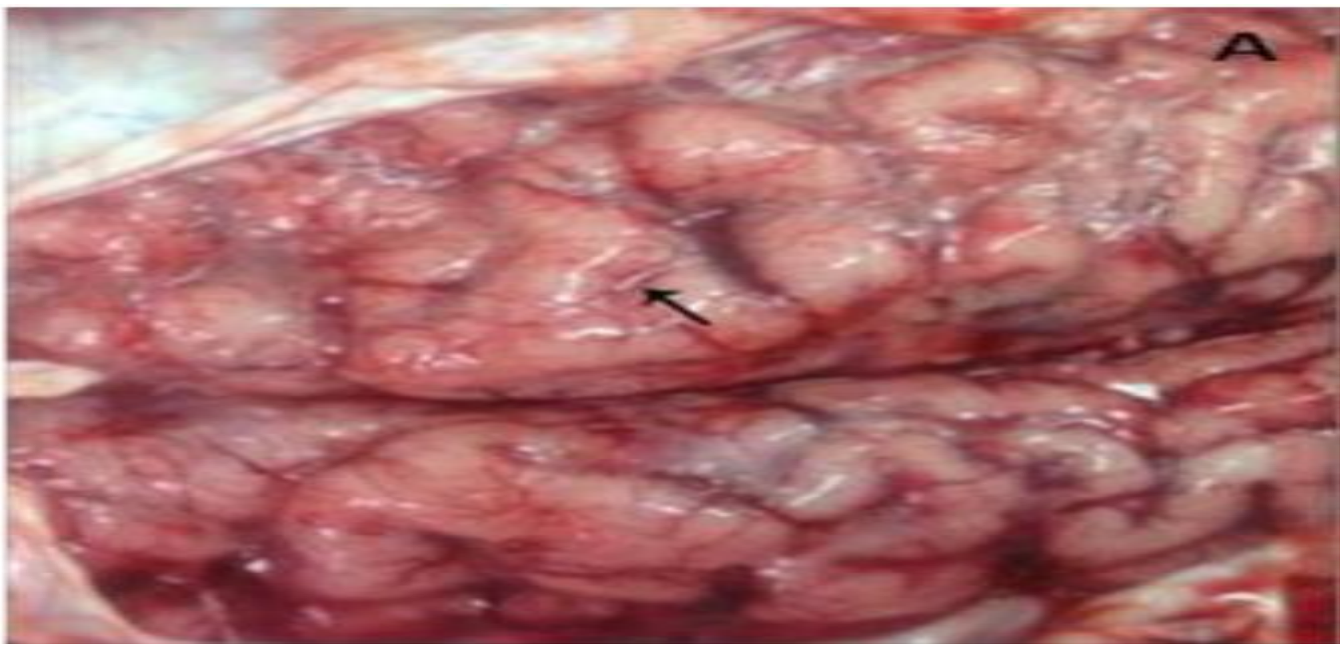
CLINICAL STAGE	Incubation period	Prodrome	Acute neurologic phase	Coma
		Fever Anorexia Headache Malaise Pain Numbness (at bite site)	Anxiety CNS signs Paralysis Hydrophobia Delirium Hallucinations	Pituitary dysfunction Hypotension Cardiac arrhythmia Cardiac arrest Coma
USUAL DURATION	20- 60 days	2-10 days	2-7 days	0-14 days

Based on: Churchill Livingstone Inc.

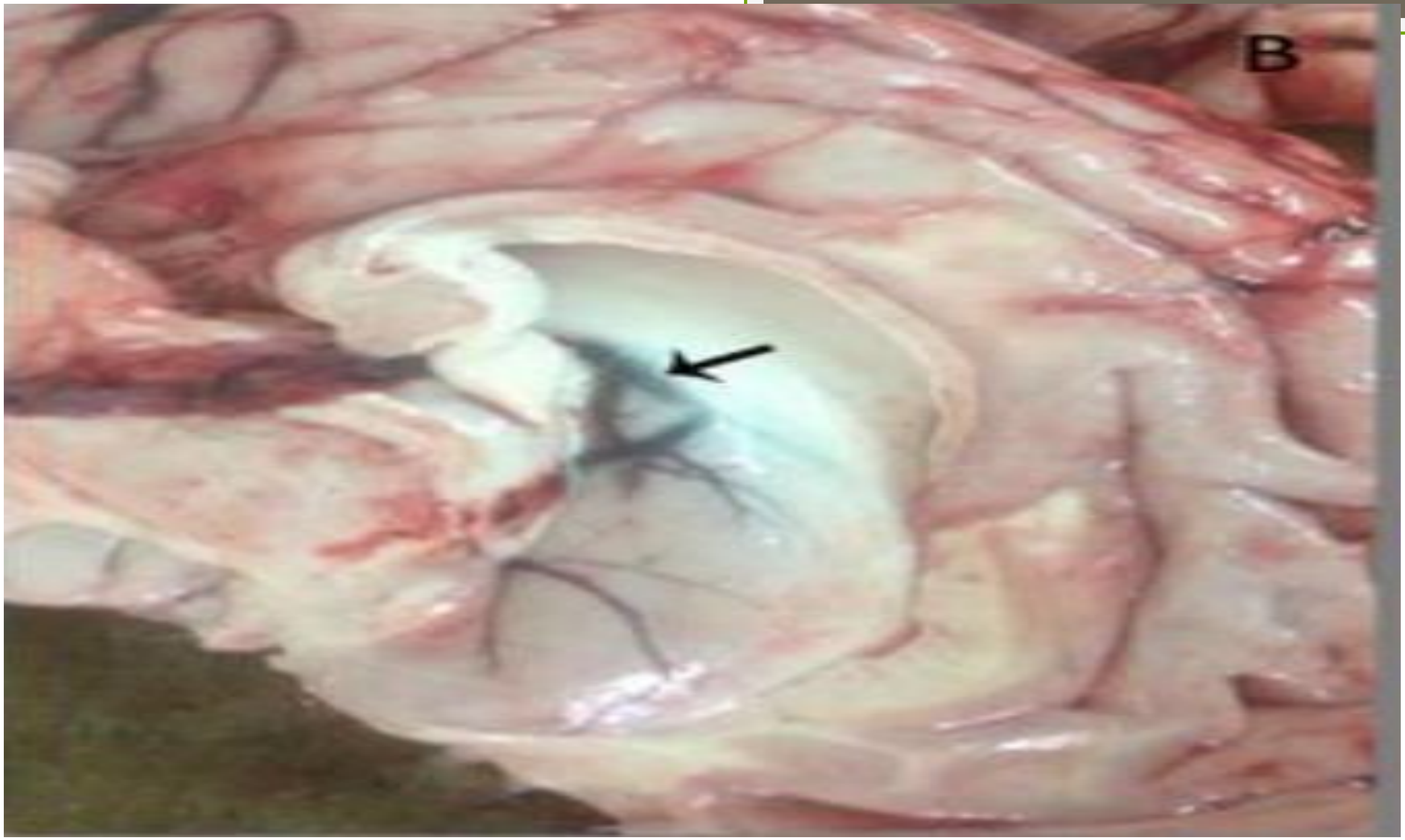
Affected man showing signs of rabies - cramps and excessive salivation (froath)

Gross examination

- No gross changes are detected in the CNS with Rabies
- In few cases post-mortem revealed severe congestion of meninges, cerebellum and cerebral hemisphere with hemorrhage.
- Edema of the cerebellum and cerebral hemisphere, in addition to, widening of gyri and narrowing of sulci of cerebral hemisphere were seen.



Congestion of meningeal blood vessels with flattening gyri and narrowing sulci (arrow);



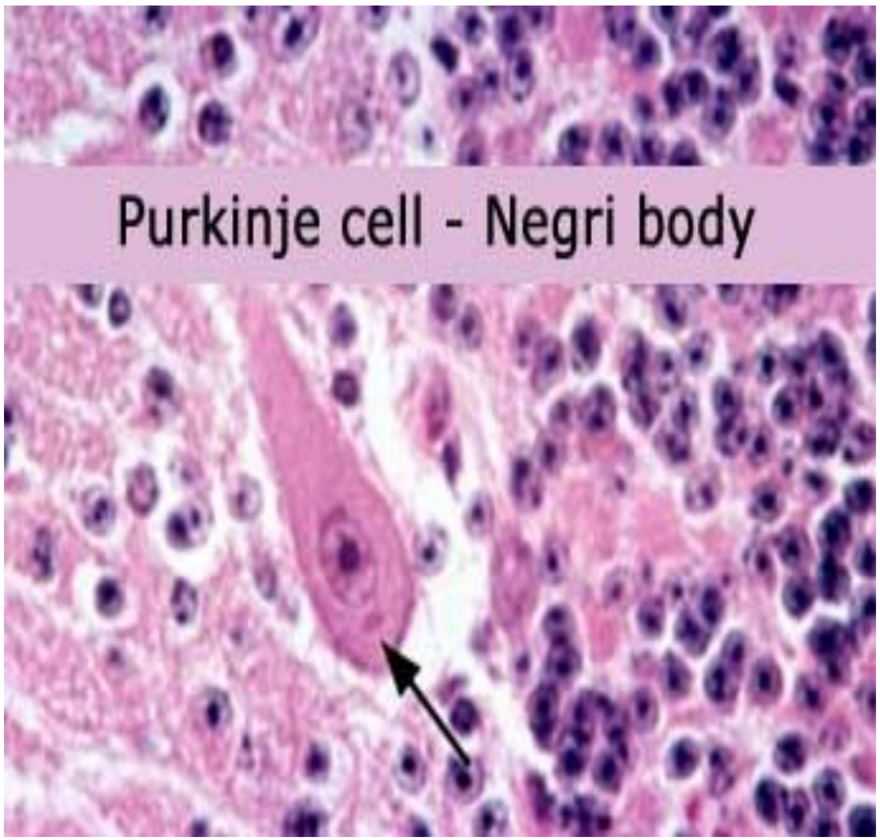
Congestion and edema in the wall of the lateral ventricle (arrow) with bulging into the lumen.

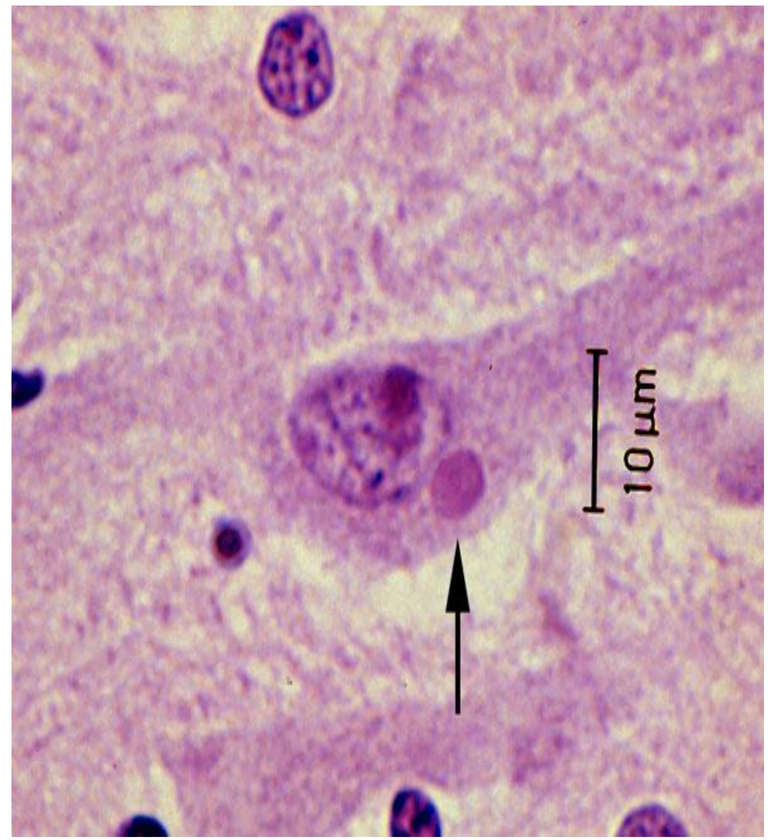
Gross picture of rabbid cow showing congestion in cerebral hemisphere

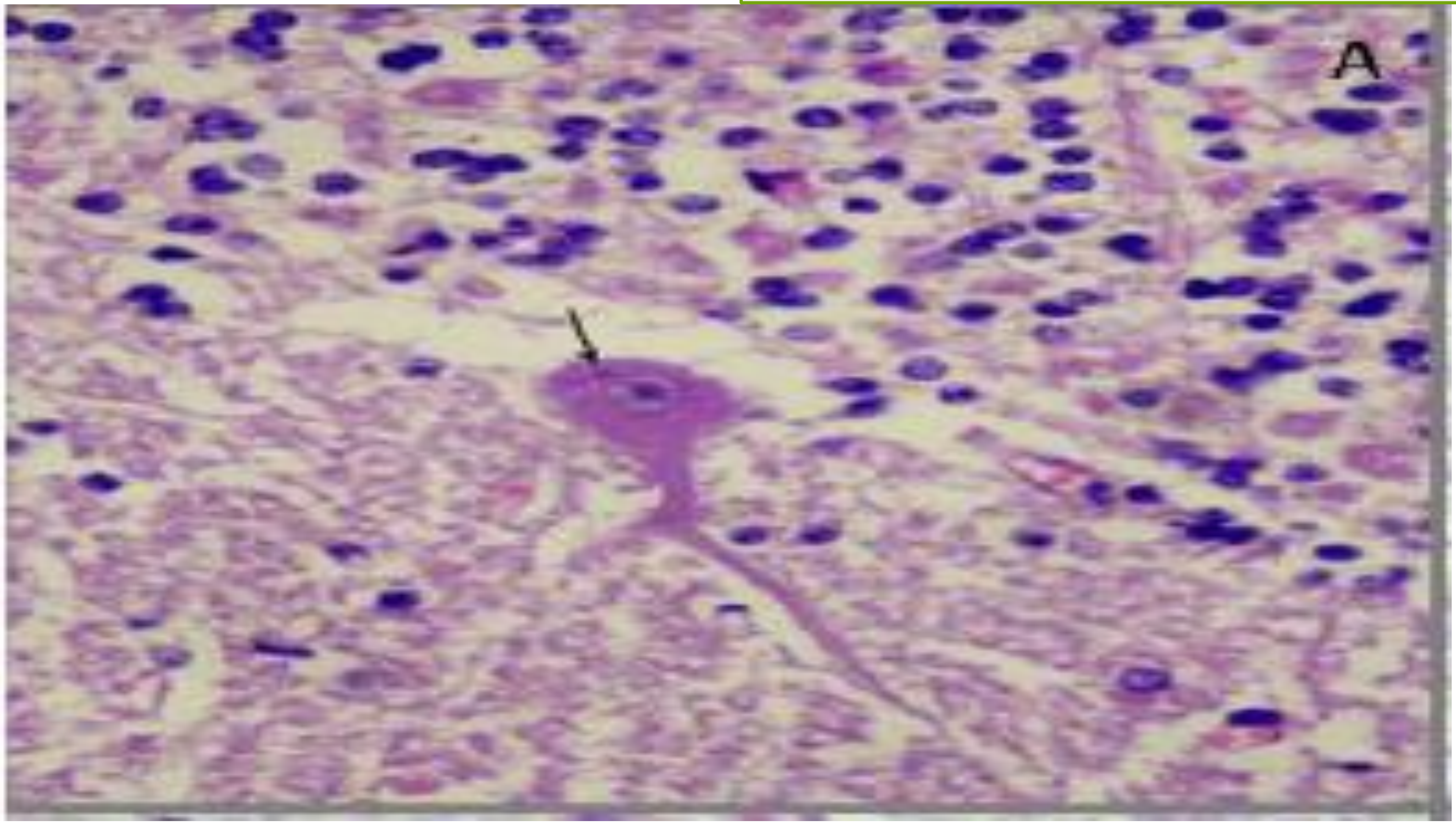
Microscopical examination

- The lesions of rabies are microscopic, limited to the central nervous system, and extremely variable in extent.
- Necrosis of neurons with specific cytoplasmic inclusion bodies in the affected nerve cells.
- Spherical cytoplasmic inclusion bodies with specific tinctorial (staining) characteristics **Negri bodies** present in neurons of dogs, cats.
- Negri bodies are not always present in rabies. Certain strains of rabies virus do not produce inclusion bodies, indicating that Negri bodies are not necessary for viral replication.

- Negri bodies are formed in only 20%(Appx.) of cases.
- In the dog, they are found mostly in the hippocampus.
- In cattle more numerous in the Purkinje cells of the cerebellum.
- In impression smears, Seller's stain is effective. The inclusion body is bright red or magenta.



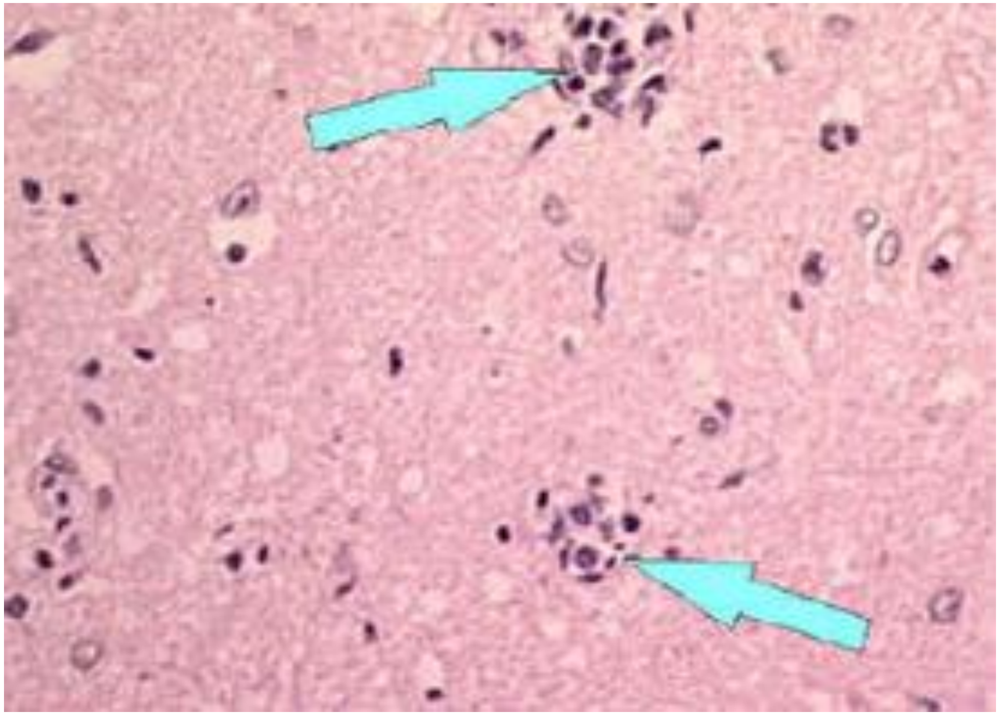




Esinophilic intracytoplasmic inclusion body (Negri body) around nucleus of Purkinje cell, (HE, x400)

- In some cases, there is **diffuse non suppurative encephalitis** characterized by perivascular cuffing, neuronophagic nodules and other changes of destruction of neurons throughout the brain. These changes are particularly prominent in the brain stem, hippocampus and the **gasserian ganglia**. (Gasserian ganglion lies on the sensory root of the trigeminal nerve, and from it originate the three branches - ophthalmic, maxillary, mandibular).

- Lesions in the gasserian ganglia consists of the collections of proliferating glial cells encroaching on the neurons and replacing them. These collections of proliferating glial cells are known as "**Babes' nodules**".



Babes nodule

Collection of samples:

- Usually the brain is collected following the opening of the skull in a necropsy room, and the appropriate samples are collected.
- This step may be hazardous if laboratory technicians are not fully trained, or under field conditions.
- In such cases, there are two possible methods of collecting some brain samples without opening the skull:

Occipital foramen route for brain sampling:
A 5 mm drinking straw or a 2 ml disposable plastic pipette is introduced into the occipital foramen in the direction of an eye. Samples can be collected from the rachidian bulb, the base of the cerebellum, hippocampus, cortex, and medulla oblongata.

Retro-orbital route for brain sampling:
In this technique, a trocar is used to make a hole in the posterior wall of the eye socket, and a plastic pipette is then introduced through this hole. The sampled parts of the brain are the same as in the former technique, but they are taken in the opposite direction.

Diagnosis of Rabies

1. Direct Microscopy: Histological Identification of Characteristic Cell Lesions

2. Demonstration of Viral Antigen

2.1. Fluorescent Antibody Technique (FAT)

2.2. Rapid Rabies Enzyme Immunodiagnosis (RREID)

3. Virus Isolation

3.1. Mouse inoculation test

3.2.Rapid tissue culture infection test(RTCT)

4. Demonstration of Antibodies

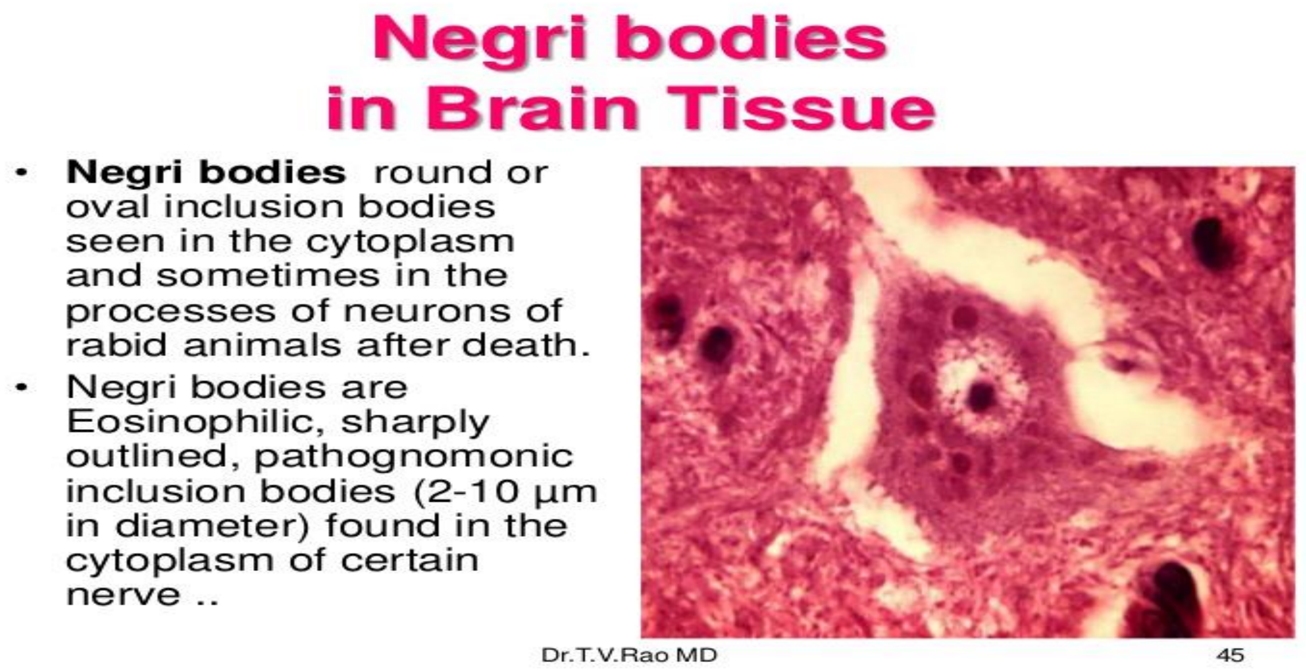
4.1. The Mouse Neutralization Test (MNT),

4.2.Rapid Fluorescent Focus Inhibition Test (RFFIT)

4.3.Fluorescence Antibody Virus Neutralization Test (FAVN)

1. Direct Microscopy: Histological Identification of Characteristic Cell Lesions

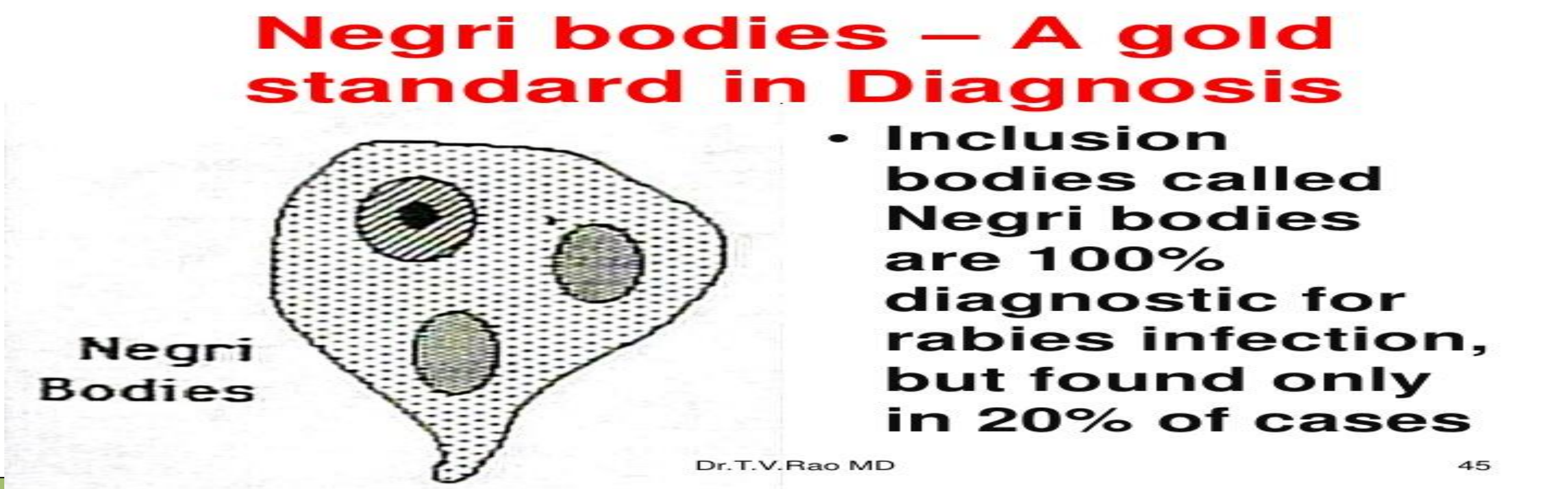
- Infected neuronal cells reveal aggregates of viral particles "**Negri bodies" intracytoplasmic inclusion bodies .**
- Demonstrated by histological tests (Seller's Technique) on smears taken from various areas of the brain.



- simple, rapid test,
- Seller’s method on unfixed tissue smears has a very low sensitivity and is only suitable for fresh specimens.

Limitation

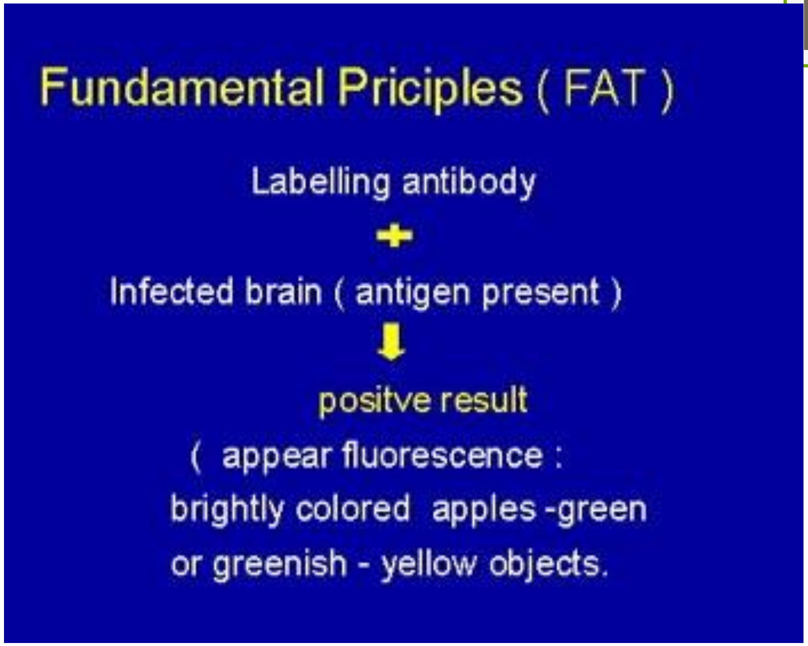
- Techniques that stain sections of paraffin embedded brain tissues are time consuming, less sensitive, and more expensive.
- Histological techniques are much less sensitive than immunological methods, especially in the case of autolysed specimens, and are no longer

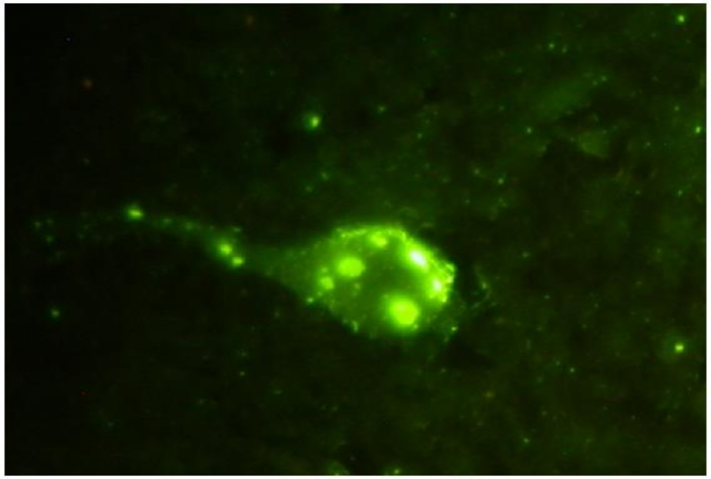


2. Demonstration of Viral Antigen

2.1 Fluorescent Antibody Technique (FAT)

- Most widely used test for postmortem rabies diagnosis
- Recommended by both World Health Organization (WHO) and World Organization for Animal Health.
- Developed by Goldwasser and Kissling in 1957
- It involves demonstration of the rabies virus nucleoprotein antigen (N) in fresh brain smears of a suspected rabies





Fluorescent antibody technique (FAT) on human brain smear positive for rabies.

- Specificity and sensitivity of the test almost approach 99%
- FAT can also be applied to specimens preserved in 50% glycerol saline after rigorous washing of the specimens with normal saline
- FAT on formalin-fixed and digested samples is always less reliable and more cumbersome than when performed on fresh tissue

- FAT can also be performed on corneal smears and nuchal skin biopsy in suspected cases, however it has been found to have limited reliability and low sensitivity for ante mortem diagnosis of rabies.

3. Virus Isolation

- Virus isolation is required for confirmatory diagnosis, especially when FAT gives an uncertain result and more importantly for molecular characterization of viruses in a geographical location and for tracing the origin of the virus if rabies occurs in a rabies-free area.

- Two techniques can be employed for this purpose

3.1. Mice inoculation technique (MIT)
3.2. Rapid tissue culture infection test (RTCT)

3.1. Mouse Inoculation Test

Three-to-ten mice, 3-4 weeks old (12–14 g), or a litter of 2-day-old newborn mice

inoculated intracerebrally with the clarified supernatant of a 10–20% (w/v) homogenate of brain material in an isotonic buffered solution containing antibiotics.

The inoculated mice are observed daily for 28 days

- they develop typical signs and symptoms of rabies any time after 5–7 days depending on the incubation period
- Further confirmation of the diagnosis can be made by extracting the brain of the diseased mouse and subjecting this to FAT.
- Disadvantage - long interval before a diagnosis can be made since the inoculated mice need to be kept under observation for 28 days as some wild viruses may have a very long incubation period.

1. Demonstration of Viral Antigen

a) Direct Rapid Immunohistochemical Test (dRIT)
b) Indirect Rapid Immunohistochemistry Test (IRIT)
c) Immunochromatographic Techniques

2. Nucleic Acid Detection Techniques

a) Reverse Transcriptase PCR (RT-PCR)
b) Real-Time PCR
c) Nucleic acid sequence-based amplification (NASBA) technique
d) Loop-Mediated Isothermal Amplification (LAMP)

Vaccination

➢Vaccination against rabies is used in two distinct situations:

❖to protect those who are at risk of exposure to rabies, i.e. preexposure vaccination;

❖to prevent the development of clinical rabies after exposure has occurred, usually following the bite of an animal suspected of having rabies, i.e. post-exposure prophylaxis.

Pre exposure vaccination-

Type of vaccine: Modern cell-culture (Human diploid cell culture) or embryonated-egg vaccine

Number of doses: Three, one on each of days 0, 7 and 21 or 28, given i/m (1 or 0.5 ml/dose depending on the vaccine) or i/d (0.1 ml/inoculation site)

Post-exposure prophylaxis.

Passive immunization- The dose for HRIG is 20 IU/kg body weight and for ERIG and F(ab')2 products 40 IU/kg body weight. The full dose of rabies immunoglobulin, or as much as is anatomically feasible, should be administered into and around the wound site.

Active immunization- Cell-culture or embryonated - egg-based rabies vaccines

The five-dose regimen is administered on days 0, 3, 7, 14 and 28 into the deltoid muscle.(I/M)

THANK YOU


WE STAY
AT WORK FOR YOU
YOU STAY
AT HOME FOR US


Take Care
and
Stay At
Home
- Covid19 -